THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · NCERT-TYPE CHAPTER TEST SET 1

Chapter 1: **Real Numbers** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The exponent of 3 in the prime factorisation of 2250 is: 1

(a) 4 **(b)** 5 **(c)** 2 **(d)** 3

2. If HCF(306, 657) = 9, then LCM(306, 657) is: 1

(a) 22338 **(b)** 963 **(c)** 2754 **(d)** 201042

3. The product of a non-zero rational and an irrational number such as $2\sqrt{3}$ is: 1

(a) always an integer **(b)** always irrational

(c) sometimes rational **(d)** always rational

4. The decimal expansion of $\frac{13}{3125}$ terminates after how many places of decimals? 1

(a) 6 **(b)** 8 **(c)** 7 **(d)** 5

5. Which of the following is irrational? 1

(a) $3 \times \sqrt{9}$ **(b)** $\sqrt{9} + 3$ **(c)** $\sqrt{25} - 3$ **(d)** $\sqrt{5} + 3$

6. For which of the following is n such that 3^n ends with the digit 0? 1

(a) $n = 2$ only **(b)** only even n

(c) every n **(d)** no natural number n

7. In Euclid's division lemma for 155 and 12, $155 = 12q + r$. The value of q is: 1

(a) 12 **(b)** 14 **(c)** 13 **(d)** 11

8. The exponent of 5 in the prime factorisation of 3600 is: 1

(a) 2 **(b)** 3 **(c)** 5 **(d)** 4

9. If HCF(510, 92) = 2, then LCM(510, 92) is: 1

(a) 1020 **(b)** 23460 **(c)** 602 **(d)** 46920

10. The product of a non-zero rational and an irrational number such as $3\sqrt{5}$ is: 1

(a) always rational **(b)** always irrational

(c) sometimes rational **(d)** always an integer

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Check whether 6^n can end with the digit 0 for any natural number n. 2

12. Given that HCF(306,657)=9, find LCM(306,657). 2

13. Explain why $7 \times 6 + 7$ is a composite number. 2

14. Check whether 14^n can end with the digit 0 for any natural number n. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Prove that $\sqrt{2}$ is irrational. 3

16. Show that $3 + 2\sqrt{5}$ is irrational, given that $\sqrt{5}$ is irrational. 3

17. Find the HCF and LCM of 26 and 91 by prime factorisation and verify HCF $\times$ LCM $=$ product. 3

OR

Prove that $\sqrt{3}$ is irrational. 3

18. Show that $2 + 3\sqrt{7}$ is irrational, given that $\sqrt{7}$ is irrational. 3

OR

Find the HCF and LCM of 510 and 92 by prime factorisation and verify HCF $\times$ LCM $=$ product. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Show that the square of any positive integer is of the form $3q$ or $3q + 1$. 5

OR

Show that any positive odd integer is of the form $6q + 1$, $6q + 3$ or $6q + 5$. 5

20. Show that the square of any positive integer is of the form $4q$ or $4q + 1$. 5

OR

Show that any positive odd integer is of the form $4q + 1$ or $4q + 3$. 5

END OF QUESTION PAPER — Total: 40 Marks

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